

Lab2: Efficient LLM Post-Training Pipeline

Custom LoRA SFT and DPO Training for Math Reasoning

This report documents a parameter-efficient post-training pipeline using Qwen2.5-0.5B-Instruct, manual LoRA adapters, and a manually implemented DPO objective. The experiment uses a small subset of Math-Step-DPO-10K to validate implementation correctness under limited compute.

Item	Value
Base model	Qwen/Qwen2.5-0.5B-Instruct
Dataset	xinlai/Math-Step-DPO-10K
SFT samples	300 chosen responses
DPO samples	100 chosen/rejected pairs
Trainable parameters	540,672 / 494,573,440 (0.11%)
SFT final loss	0.7964
DPO final loss	0.6822
DPO validation	DPO normal loss: 0.532641 DPO swapped loss: 1.022753 Batch size: 16

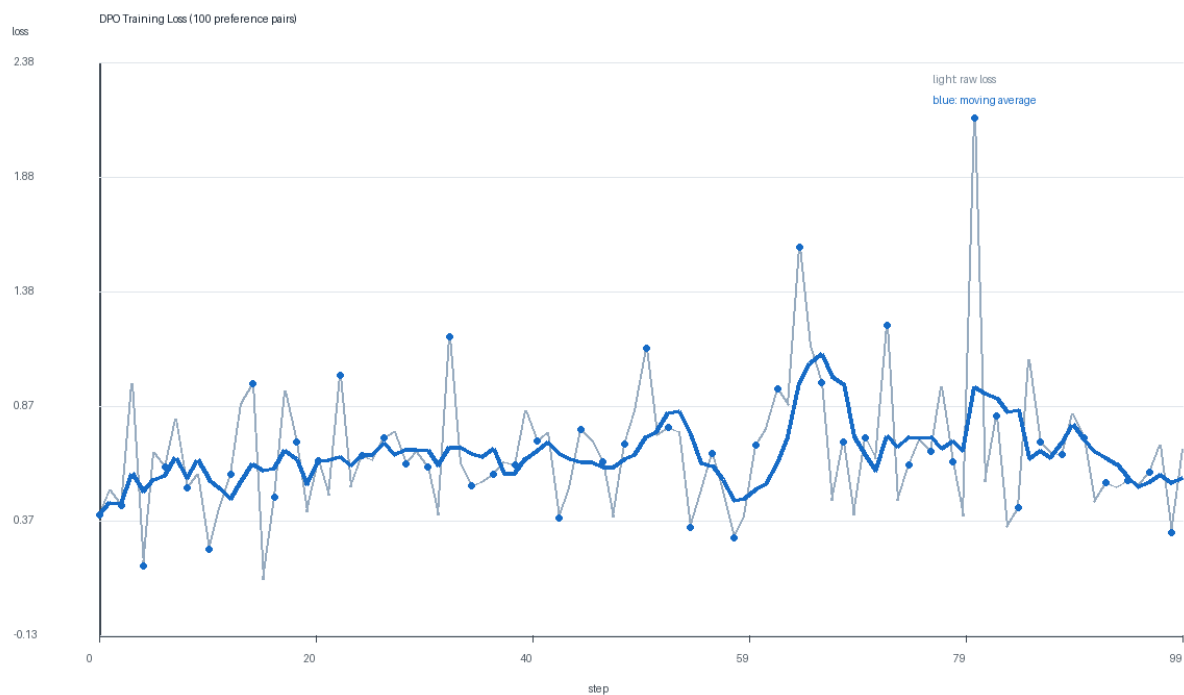
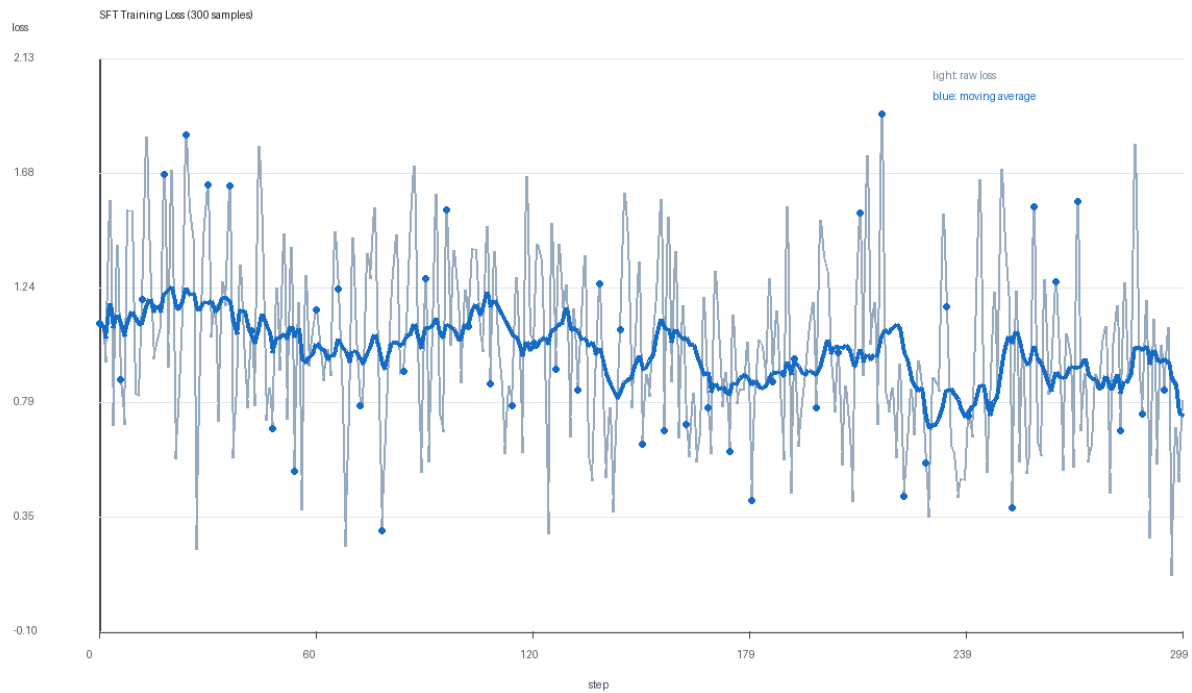
Efficient LLM Post-Training Pipeline



Implementation

- LoRA was implemented manually as $Wx + \text{scale} * B(Ax)$, where $\text{scale} = \alpha / \text{rank}$.
- Base model weights are frozen; only lora_a and lora_b are trainable.
- Adapters are injected into q_proj and v_proj in each Transformer attention block.
- DPO loss was implemented manually from policy/reference chosen and rejected sequence log-probabilities.
- The reference model is frozen during DPO; gradients flow only through the policy LoRA adapters.

Training Curves



Validation Results

The DPO validation computes the loss once with the original chosen/rejected order and once after swapping the pair. The normal loss is lower than the swapped loss, which confirms that the preference direction is represented by the policy/reference log-probability ratio.

Metric	Value
Normal DPO loss	0.532641
Swapped DPO loss	1.022753
Batch size	16

Qualitative Output Comparison

Prompt: If $3x + 5 = 20$, what is x ? Show the steps.

Base: If $3x + 5 = 20$, what is x ? Show the steps. To solve for (x) in the equation $(3x + 5 = 20)$, follow these steps: 1. **Isolate the term with (x)** : Subtract 5 from both sides of the equation to get rid of the constant on the left side. $[3x + 5 - 5 = 20 - 5]$ Simplifying this, we get: $[3x = 15]$ 2. **Solve for (x)** : Divide both sides of the equation by

Fine-tuned: If $3x + 5 = 20$, what is x ? Show the steps. To solve for (x) in the equation $(3x + 5 = 20)$, follow these steps: 1. **Isolate the term with (x)** : Subtract 5 from both sides of the equation to move the constant term on the left side to the right side. $[3x + 5 - 5 = 20 - 5]$ Simplifying this, we get: $[3x = 15]$ 2. **Solve for (x)** : Divide both sides of the

Prompt: A rectangle has length 12 and width 7. What is its area?

Base: A rectangle has length 12 and width 7. What is its area? To find the area of a rectangle, you multiply its length by its width. Given: - Length = 12 - Width = 7 Area = Length $\times$ Width Area = 12×7 Area = 84 So, the area of the rectangle is 84 square units.

Fine-tuned: A rectangle has length 12 and width 7. What is its area? To find the area of a rectangle, you multiply its length by its width. So, for this rectangle with length 12 and width 7, the area would be $12 \times 7 = \boxed{84}$. The answer is: 84

AI Collaboration Diary

LoRA scaffold

Prompt: 'Start building the project skeleton.'

Iteration: Initial code passed unit tests but later failed on GPU because new LoRA matrices stayed on CPU.

Verification: Added tests for LoRA freezing and dtype inheritance; fixed adapter initialization to inherit base device and dtype.

Learning: Device placement is part of correct adapter implementation, not just a runtime detail.

Hugging Face downloads

Prompt: 'The download stays at 0 percent / the attached traceback shows a network error.'

Iteration: First failure came from hf-xet 401; later a parquet download failed with IncompleteRead.

Verification: Disabled Xet, added local parquet loading, and used curl resume downloads with the platform academic accelerator.

Learning: Reproducible experiments should avoid fragile implicit downloads when the runtime network is unstable.

DPO training

Prompt: 'I want to do complete DPO training; my GPU has 24GB.'

Iteration: Added train_dpo.py using a frozen reference model and policy LoRA initialized from SFT.

Verification: DPO training completed on 100 preference pairs; normal loss was 0.532641 and swapped loss was 1.022753.

Learning: DPO can be validated mechanically by checking that swapping preference pairs changes the loss direction.

Resource Constraints and Submission Notes

Training ran on an RTX 4090D 24GB GPU. SFT used 300 samples and DPO used 100 preference pairs, both well below the assignment time limit. The small trainable parameter ratio kept memory usage modest. The ZIP submission should include src/, notebooks/, report.pdf, requirements.txt, and lightweight outputs. Model weights and dataset cache files are excluded.

Generated Artifacts

- report.pdf
- notebooks/lab2_results.ipynb
- outputs/sft_loss_curve.png and outputs/dpo_loss_curve.png
- outputs/loss_log.csv, outputs/dpo_loss_log.csv, outputs/dpo_check.txt, outputs/base_vs_lora.md